

CS 486/686

Independence and Bayesian Networks I

Yuntian Deng

Lecture 7

RN 12.4, 13.1–13.2 · PM 8.2–8.3

Reminders

DUE TONIGHT

Tue Jun 2, 11:59 PM

Chat 3 (Probabilities) on Chrysalis.

Search ›

Uncertainty ›

Decisions ›

Learning

Learning goals

- Tell if two variables are **independent**, or **conditionally independent** given a third
- Derive a **compact representation** of a joint distribution using independence
- Describe the components of a **Bayesian network**
- Compute joint probabilities *from* a Bayes net
- Explain independence in the three key structures

Why Bayes nets? The 2^n problem

A joint distribution over n Boolean variables needs 2^n probabilities.

Variables	Probabilities
6 (Holmes)	64
10	1,024
20	~ 1 million
30	~ 1 billion

Independence lets us factor this huge joint into tiny pieces.

Unconditional independence

Independence. X and Y are (*unconditionally*) independent iff (any, and therefore all):

$$P(X|Y) = P(X)$$

$$P(Y|X) = P(Y)$$

$$P(X \wedge Y) = P(X) P(Y)$$

Learning Y does **not** change our belief about X .

Two probabilities (the priors) suffice to recover four joint probabilities.

Conditional independence

Conditional independence. X and Y are *conditionally independent* given Z iff (any, and therefore all):

$$P(X|Y \wedge Z) = P(X|Z)$$

$$P(Y|X \wedge Z) = P(Y|Z)$$

$$P(X \wedge Y|Z) = P(X|Z) P(Y|Z)$$

Once we know Z , learning Y doesn't change our belief about X .

Important: independence does *not* imply conditional independence, and vice versa. We'll see both on later slides.

Quiz pack: compact representation

Three Boolean variables A, B, C . How many probabilities specify the joint?

Q1. No independence assumed.

7. Chain rule: $P(A) + P(B|A) + P(C|A \wedge B) = 1 + 2 + 4 = 7$.

Q2. A, B, C all mutually independent.

3. Just the priors: $P(A), P(B), P(C) = 3$.

Q3. A, B conditionally independent given C .

5. $P(C) + P(A|C) + P(B|C) = 1 + 2 + 2 = 5$.

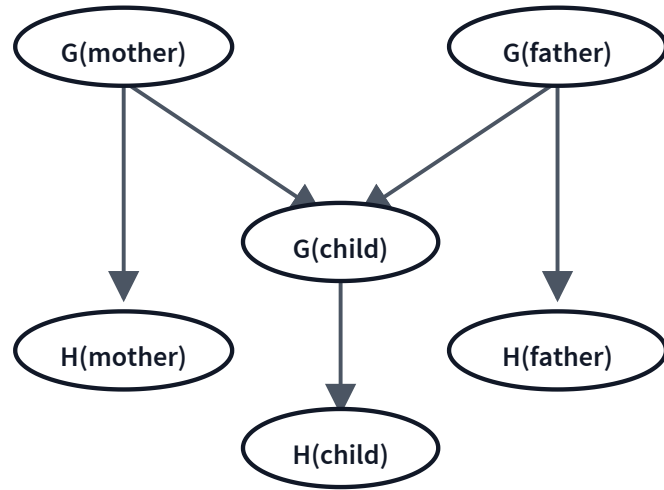
Bayes net = DAG + CPTs

A **Bayesian network** is a *directed acyclic graph* (DAG) in which:

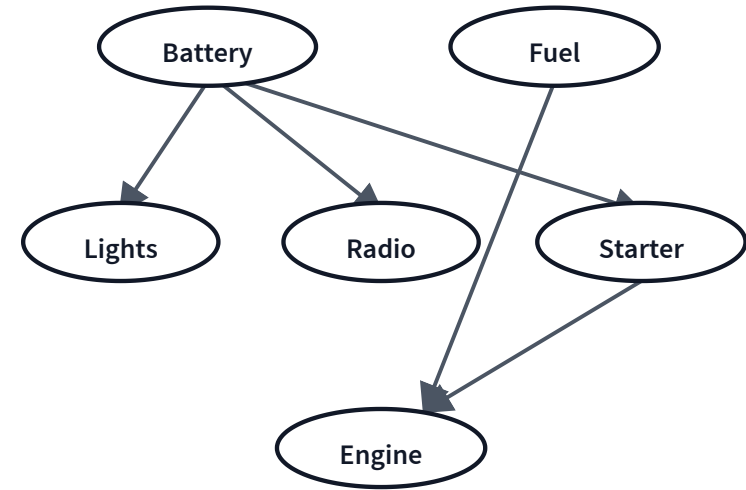
- each node is a random variable,
- each edge $X \rightarrow Y$ says " X directly affects Y " — X is a *parent* of Y ,
- each node X_i carries a conditional probability table $P(X_i \mid \text{Parents}(X_i))$.

Examples of Bayes nets

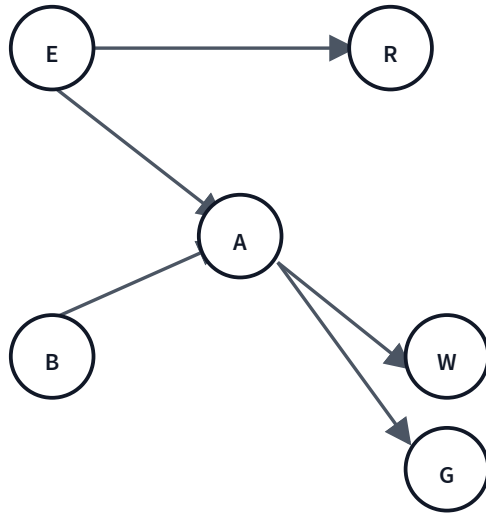
Inheritance of handedness



Car diagnostic network



The Holmes Bayes net



$$P(E) = 0.0003$$

$$P(B) = 0.0001$$

$$P(R \mid \cdot)$$

$$E: 0.9 \quad \neg E: 0.0002$$

$$P(A \mid \cdot)$$

$$\begin{array}{ll} B \wedge E: 0.96 & B \wedge \neg E: 0.95 \\ \neg B \wedge E: 0.20 & \neg B \wedge \neg E: 0.01 \end{array}$$

$$P(W \mid \cdot)$$

$$A: 0.80 \quad \neg A: 0.40$$

$$P(G \mid \cdot)$$

$$A: 0.40 \quad \neg A: 0.04$$

From 64 numbers to 12

Representation	Probabilities needed
Full joint over 6 Boolean variables	$2^6 = 64$
Holmes Bayes net: $1 + 1 + 2 + 4 + 2 + 2$	12

From those 12 numbers we can reconstruct every one of the 64 joint probabilities.

For larger networks, the savings are exponential.

Joint = product of conditionals

Bayes-net factorization. For a Bayes net over variables $X_1, \dots, X_n$:

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i)).$$

Each variable is conditionally independent of its non-descendants given its parents.

Worked example: a joint from the Holmes BN

Goal: $P(\neg B \wedge \neg E \wedge A \wedge \neg R \wedge G \wedge W)$. All neighbours call but neither a burglary nor an earthquake actually happened.

Step 1. Factor by the BN structure (using parents):

$$= P(\neg B) \cdot P(\neg E) \cdot P(A|\neg B \wedge \neg E) \cdot P(\neg R|\neg E) \cdot P(G|A) \cdot P(W|A)$$

Step 2. Plug in numbers:

$$\begin{aligned} &= 0.9999 \times 0.9997 \times 0.01 \times 0.9998 \times 0.40 \times 0.80 \\ &\approx \mathbf{3.2 \times 10^{-3}} \end{aligned}$$

Q: a different all-negatives joint

Q. What is $P(\neg B \wedge \neg E \wedge \neg A \wedge \neg R \wedge \neg G \wedge \neg W)$?

- A. 0.5699
- B. 0.6699
- C. 0.7699
- D. 0.8699
- E. 0.9699

A — 0.5699. By the same factorization, each factor is now a "no event happened" version:
 $= 0.9999 \times 0.9997 \times (1 - 0.01) \times (1 - 0.0002) \times (1 - 0.04) \times (1 - 0.40) \approx 0.5699.$

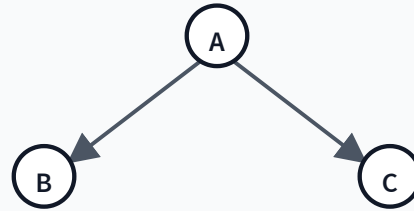
Three key structures

Chain



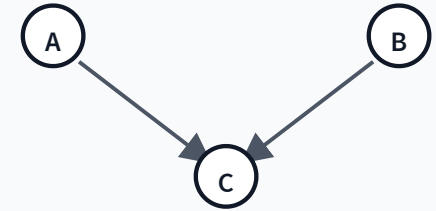
A and C talk through B.

Common cause



B and C share a hidden cause A.

Common effect



A and B both cause C (a v-structure).

Each structure has its own independence pattern. Let's check.

Chain: $B \rightarrow A \rightarrow W$



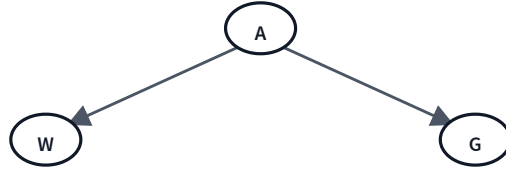
Q1. Are B and W *unconditionally* independent?

No. Knowing B makes the alarm more likely, which makes W more likely. Information flows through A .

Q2. Are B and W *conditionally* independent given A ?

Yes. Once we know whether the alarm is going, W only depends on A ; B adds nothing more. *The middle node "blocks" the path.*

Common cause: $W \leftarrow A \rightarrow G$



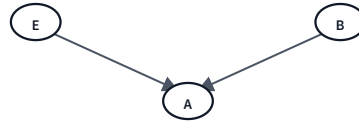
Q1. Are W and G *unconditionally* independent?

No. If W is calling, the alarm is more likely going, so G is more likely too. The shared cause A couples them.

Q2. Are W and G *conditionally* independent given A ?

Yes. Once A is known, the two callers are independent observers. *The common cause "blocks" the path when conditioned.*

Common effect: $E \rightarrow A \leftarrow B$



Q1. Are E and B *unconditionally* independent?

Yes. Earthquakes and burglaries don't influence each other — they only share the same downstream alarm.

Q2. Are E and B *conditionally* independent given A ?

No — the opposite. Given the alarm, learning of an earthquake makes a burglary less likely: **explaining away**. Conditioning on the common effect *creates* a dependence.

The v-structure is the surprise: conditioning on the middle node *opens* a path instead of blocking it.

The three rules

Structure	Are endpoints independent?	Given the middle node?
Chain $A \rightarrow B \rightarrow C$	No	Yes <i>blocked</i>
Common cause $B \leftarrow A \rightarrow C$	No	Yes <i>blocked</i>
Common effect $A \rightarrow C \leftarrow B$	Yes	No <i>opens up</i>

In the chain and common-cause structures, knowing the middle node *blocks* information flow. In the v-structure, knowing the middle node *opens* it (explaining away).

Learning goals (recap)

- ✓ Tell if two variables are independent or conditionally independent
- ✓ Derive a compact representation using independence
- ✓ Describe the components of a Bayesian network
- ✓ Compute joint probabilities from a Bayes net
- ✓ Explain independence in the three key structures

Next: d-separation

The three structures cover paths of length 2. For arbitrary paths in a Bayes net, we generalize to **d-separation** — the rule for reading off all (conditional) independences from any DAG.